

Research Interest

My research interests lie at the intersection of deep learning, medical image analysis, multimodal learning, computer vision, and trustworthy AI. My broader goal is to develop learning systems that can extract and integrate clinically meaningful information from complex biomedical data while preserving subtle pathological characteristics. I am particularly interested in models that are accurate, interpretable, computationally efficient, and capable of learning meaningful relationships among multiple biomarkers and data sources.

My research journey began with medical image classification and segmentation. During my undergraduate thesis, I worked on multiclass brain tumor classification and segmentation from 2D MRI images, where I developed a lightweight custom CNN for classification and implemented a Residual Attention U-Net for tumor segmentation. This work introduced me to medical image preprocessing, feature extraction, localization of small pathological regions, and the importance of preserving diagnostically relevant information during image processing.

Building on this experience, my research expanded to retinal fundus imaging, neurological MRI, histopathology, and structured healthcare data. Across these projects, I worked with pretrained CNN architectures, attention mechanisms, ensemble learning, explainable AI, and specialized preprocessing techniques. I developed an eleven-step retinal preprocessing framework to enhance physiologically distinct vessel and lesion information and also explored a Gamma-CLAHE-based preprocessing approach for neurological MRI analysis. These experiences gradually shifted my interest from simply improving classification accuracy toward understanding what information a model learns, how pathological features are represented, and how different visual biomarkers interact during prediction.

This direction is central to my current MSc thesis on pathology-preserving lesion–vessel interaction learning for diabetic retinopathy severity grading. Rather than treating retinal lesions and vascular abnormalities as independent features, I am investigating their structural and spatial relationships across different stages of diabetic retinopathy. My objective is to preserve subtle pathological information while learning disease-related lesion–vessel interactions and progression patterns. I am also interested in transferring this knowledge to a lightweight model through knowledge distillation, allowing diagnostically important representations to be retained with lower computational requirements.

This work has naturally led me toward multimodal and relational representation learning. Biomedical problems often contain complementary information across anatomical structures, imaging modalities, physiological signals, biomarkers, and clinical variables. I am interested in studying how these heterogeneous sources can be effectively integrated using attention-based fusion, cross-modal learning, graph-based methods, and transformer architectures. My focus is particularly on understanding the relationships among different sources of information rather than processing them independently.

Alongside multimodal learning, I am interested in developing generalizable and trustworthy AI systems. Medical datasets often vary across institutions, imaging devices, patient populations, and acquisition conditions, while rare diseases and advanced disease stages may contain limited training samples. I am therefore interested in data-efficient learning, generative approaches for clinically meaningful synthetic data, explainable AI, and privacy-preserving methods such as federated learning. These directions can help improve model robustness while maintaining interpretability and protecting sensitive patient information. More recently, my interests have expanded toward foundation models and scientific machine learning. I am particularly interested in understanding how large pretrained models learn transferable representations, adapt to specialized scientific domains, and integrate multimodal information. I would like to investigate how the principles of foundation models can be applied to biomedical and scientific data, where limited annotations, heterogeneous modalities, and domain-specific knowledge remain important challenges.

Overall, my research direction centers on developing intelligent, trustworthy, and scientifically meaningful learning systems for complex multimodal data. I am interested not only in achieving strong predictive performance, but also in understanding how models learn representations, how different biomarkers and modalities interact, and how clinically important information can be preserved during learning.